Project Documentation

Project Title: LibSpace - A Robust Backend for Library Operations

1. Introduction

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Team ID: SWTID-2026-6809

Team Leader:

S. Ayesha Sulaiha (ayeshasulaiha1017@gmail.com)

Team Members:

1) S. Abinaya (abinayaabi0385@gmail.com)

2) S. Keerthika (keerthisanmugam67@gmail.com)

3) V. Swetha (swethavishwanathan42@gmail.com)

4) A. Rishiyana Begam (rishiyanabegam8@gmail.com)

2. Project Overview

Purpose:

The purpose of LibSpace is to develop a backend system that manages library book records and user accounts securely.

It demonstrates user authentication, secure API development, database integration, and CRUD operations.

Features:

- **User registration and login**
- **Secure password hashing using bcrypt**
- **JWT-based authentication**
- **Add, update, and delete books**

- View book list and get book by ID

3. Architecture

API Layer (Routes): Handles incoming HTTP requests and API endpoints using Node.js and Express.js.

Routes:

/api/auth/register

/api/auth/login

/api/book/add

/api/book/:id

Business Logic Layer (Controllers):

Handles authentication, password hashing, JWT generation and CRUD operations.

Data Access Layer (Database/Models):

Handles MongoDB database interaction using Mongoose.

Models represent Users and Books collections.

4. Technologies Used

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT (JSON Web Token)
- bcrypt
- Thunder Client
- Visual Studio Code

5. Installation

Install Dependencies:

npm install

Environment Variables (.env):

PORT=5000

MONGO_URI=mongodb://127.0.0.1:27017/libspace

JWT_SECRET=your_secret_key

Run the Server:

npm start

or

npm run dev

6. Folder Structure

Libspace/

■■■ config/

■■■ controllers/

■■■ models/

■■■ routes/

■■■ middleware/

■■■ server.js

■■■ package.json

7. API Documentation

Users APIs:

POST /api/auth/register

POST /api/auth/login

DELETE /api/auth/delete

POST /api/auth/logout

Book APIs:

POST /api/book/add

GET /api/book

GET /api/book/:id

PUT /api/book/:id

DELETE /api/book/:id

8. Authentication

LibSpace uses JWT for authentication. After login, the server generates a token

which must be sent in request headers for protected routes.

Authorization: Bearer

9. Database Design

Users Collection Fields:

name

email

password

createdAt

updatedAt

Books Collection Fields:

title

author

url

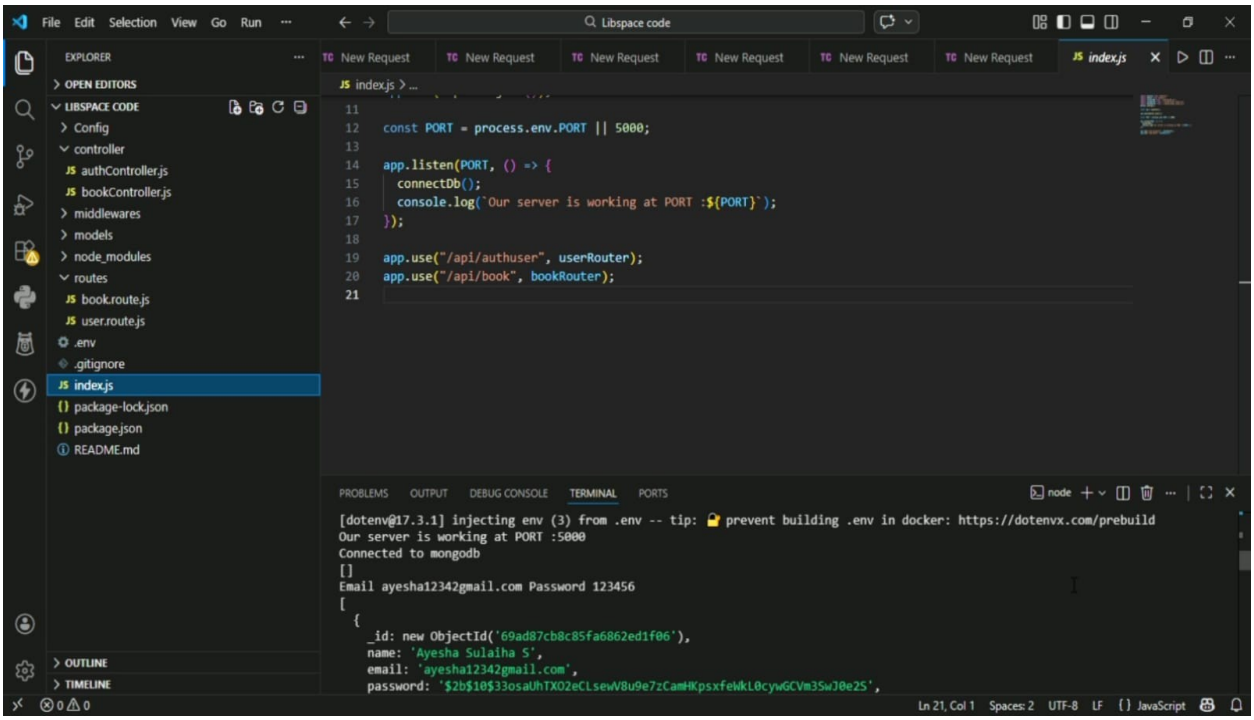
user (Object reference)

10. Testing

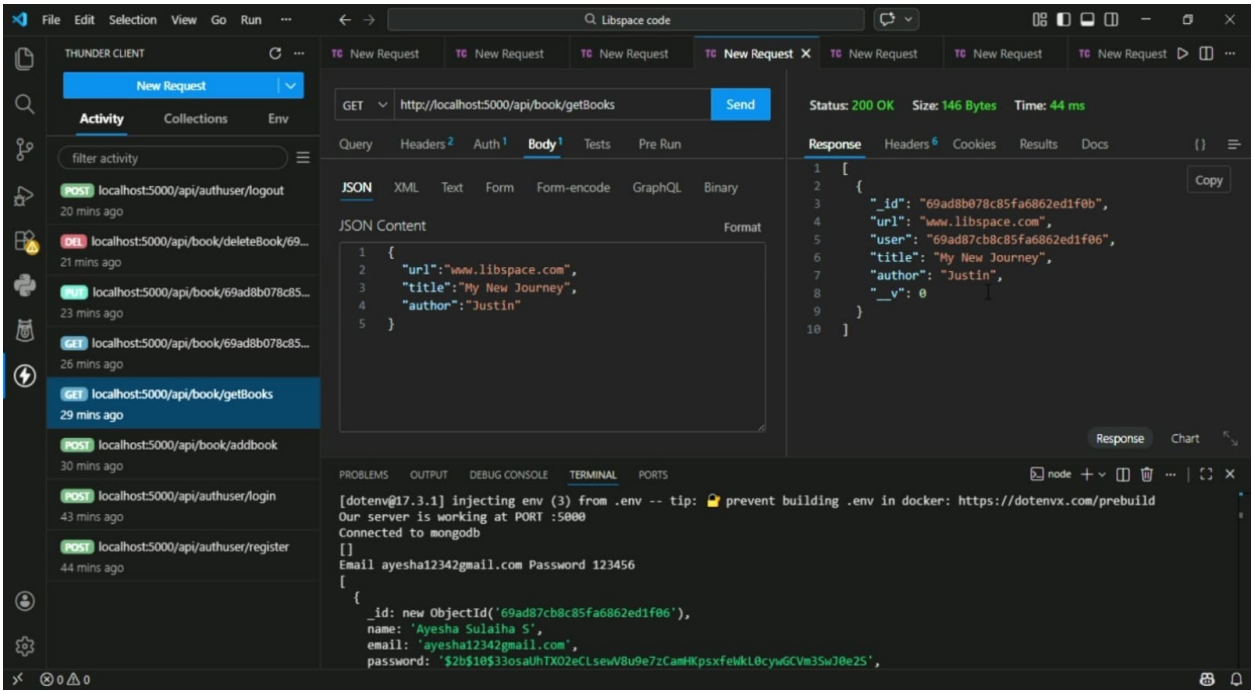
1. Tested using Thunder Client
2. Verified registration and login
3. Verified book CRUD operations
4. Verified token authentication

11. Screenshots

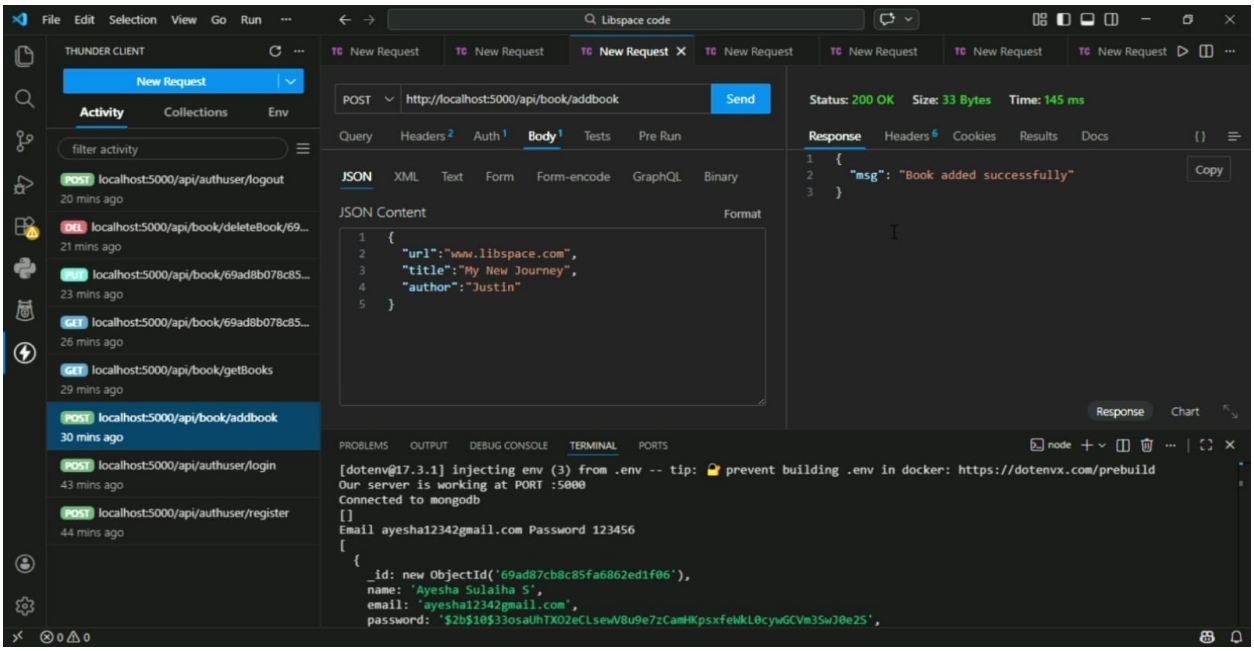
Successful Connection to MongoDB



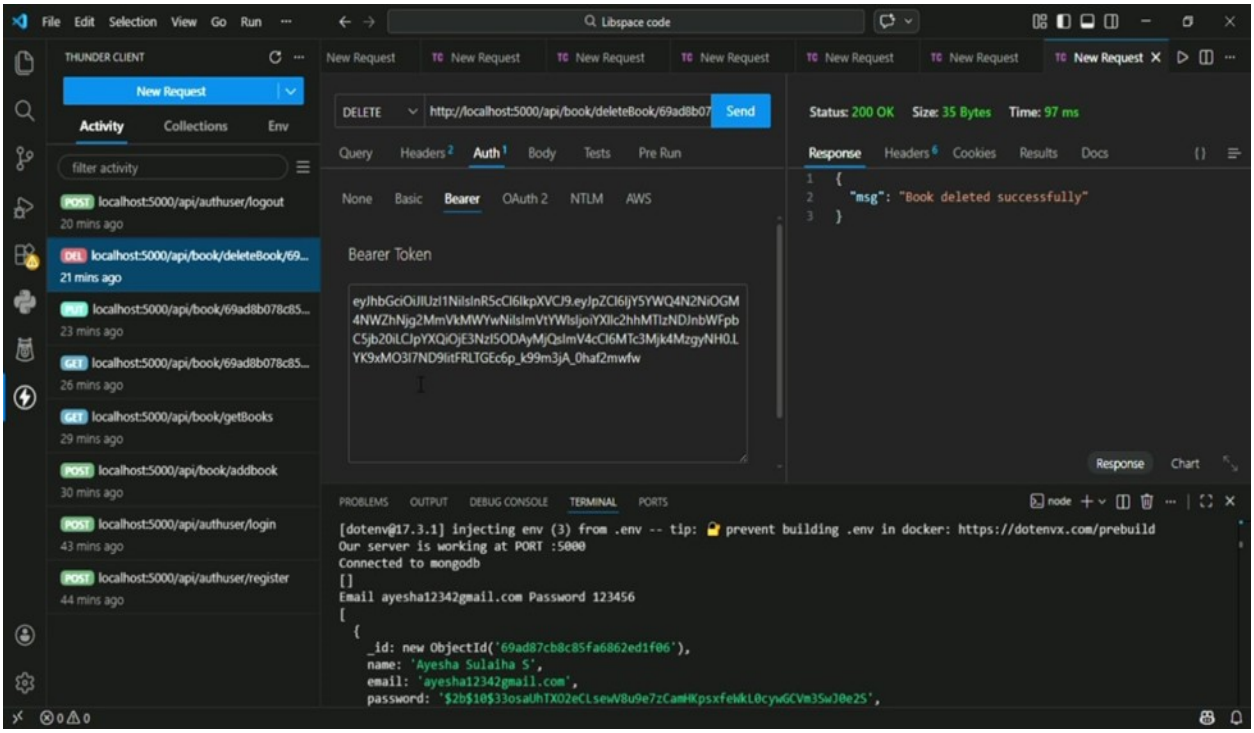
Displaying Book Title and Author



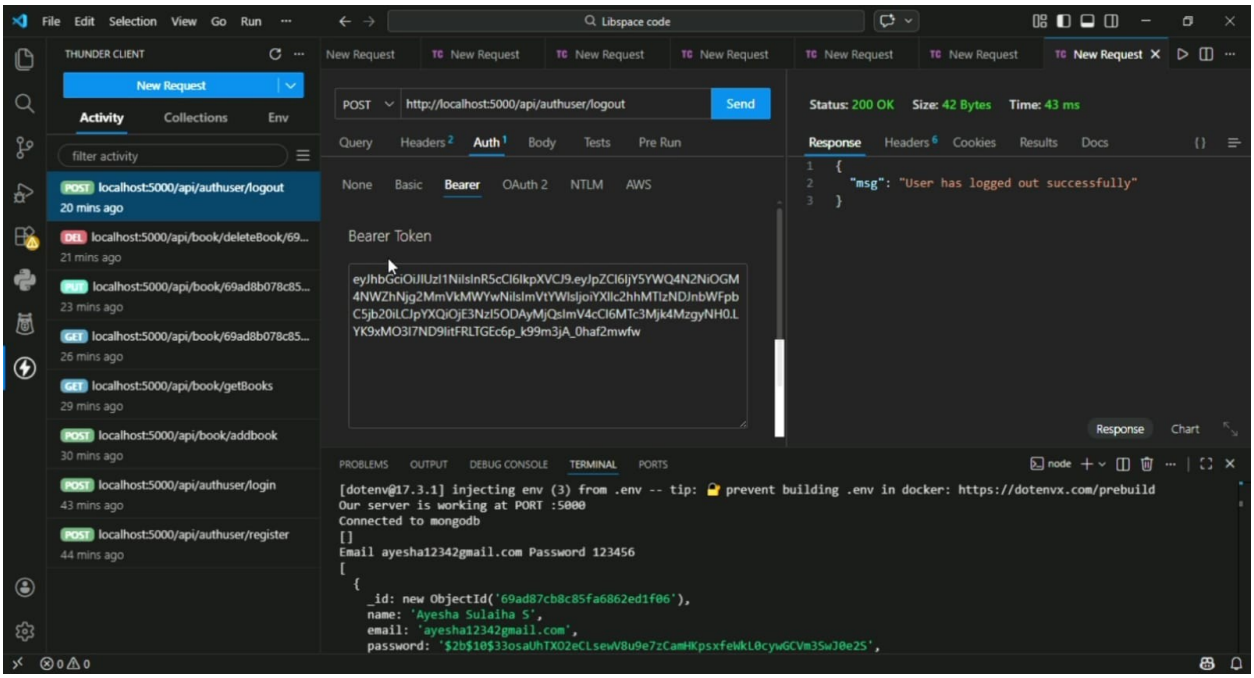
Adding a Book



Deleting a Book



User Logout Response



12. Known Issues

- **No frontend interface**
- **Requires Thunder Client for testing**
- **Token blacklist optimization needed**

13. Future Enhancements

- Frontend integration with React
- Role-based authentication
- Cloud database deployment
- Book file upload support
- Mobile application support

14. Demo Video

https://drive.google.com/file/d/1-7RYgoBT1L1dOpl--W3i4xr88a-DguYP/view?usp=sharing

15. Libspace Project Code

https://drive.google.com/drive/folders/1_bYY6CaN1uWB6pdFYpwfyJqSS0CgI-MO